

**A Framework for Investor Profiling and
Personalized Recommendations Integrating Risk,
ESG, and Market Preferences**

Project ID – 25-26J-535

Project Proposal Report

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Declaration

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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Date

Abstract

The proposed study will create an Investor Profiling and Recommendation System based on low-code/no-code technologies that will help to make decisions in the international capital markets. Investors now have an excessive variety of choices in the stock, commodity, index, and fund sectors, in addition to the variables of risk aversion, investment duration, geographic orientation, and Environmental, Social, and Governance (ESG) orientation. The proposed system will offer a platform that investor needs and produces the personalized recommendations of the investment in a clear and transparent way by using a carefully filtered dataset of about 400 instruments and a matching engine based on rules. The research will also look into the shortcomings of accessibility and affordability regarding use of available advisory tools which in most cases mandate expensive coding and infrastructure that smaller companies can not maintain. Through the ability to show the possibility of creating such systems based on low-code/no-code platforms, the research is anticipated to offer a cost-effective, scalable, and workable solution to advisory firms and investors. It is going to play a role in enhancing the inclusivity and transparency of the financial advisory services, which will eventually lead to investing more trust by the investor, as well as contributing to the improvement of decision-making in global markets.

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1. Introduction

The global capital market has become a competitive and diversified market as it presents the investor with a large variety of instruments such as equities, commodities, indices, and funds. This variety has generated a possibility of better portfolio diversification and expansion; however, it has also complicated the decision-making process.

Investors are now anticipated to look beyond financial returns and to focus on their risk levels, geographic risk, industry concentration, and the ethics of the business like Environmental, Social, and Governance (ESG) business aspects. The conventional techniques of advisory, based upon manual profiling and subjective decision making, are becoming ineffective in addressing the various multidimensional needs.

Robo-advisors and other automated platforms are partial solutions, and usually demand extensive amounts of code, expensive infrastructure, or proprietary data. Smaller advisory firms that comprise a big percentage of the financial services industry are often incapable of implementing such tools. A possible way out of this dilemma is low-code/no-code technologies. Such platforms allow the development of functional systems that do not require a lot of programming and lower the threshold to digital innovation.

Such technologies can be applied to the investor profiling and recommendation systems to enable advisory firms to deliver an efficient and transparent and personalized services at a reduced cost.

1.1 Background

Internationalization of financial markets has provided investors with a chance of interacting with international instruments other than their national economies. International capital movement has increased access to stocks, commodities and indices making the investment environment more interdependent. In spite of this having delivered diversification benefits, it has also complicated the process of choosing the right investment options.

Investor expectations have changed alongside the process of globalization. There is a rising demand among people and organizations to have portfolios, which are sustainable and ethical. ESG requirements have added another layer to the investment decision-making process that necessitates investigation beyond the conventional returns-risk models.

With the small and medium-size advisory firms, it is difficult to meet these changing demands. These firms tend to rely on manual techniques as opposed to other large companies that have access to proprietary tools and internal development departments that are susceptible to inefficiency and inconsistencies.

The recent development of low-code/no-code platforms represents one of the ways to eliminate these obstacles. With such technologies, complex processes can be automated and systemized without intense programming and are thus available to firms with limited technical resources. The use in investment advisory gives them the chance to create systems that are scalable, transparent and personalized.

1.2 Literature Survey

Portfolio theory and financial advisory have been heavily concerned with investor profiling. The model of portfolio selection that was developed by Markowitz [1] formed the basis of risk and return balance, and the further investigations have been able to point at the role played by the risk tolerance assessment as the guideline to the investment decision [2]. The traditional form of profiling, however, is usually based on questionnaires that are not dynamic and thus, it might not reflect dynamic investor preferences.

These limitations were tried to be solved with the introduction of robo-advisory platforms that automated processes of profiling and recommendations. It has been established that robo-advisors can enhance the effectiveness and availability of financial services [3]. However, most robo-advisors are domestic market-oriented and need a considerable amount of coding and technical support [4]. This places obstacles to small and medium advisory firms which are unable to invest in the development resources.

Concomitantly, the role of incorporation of the Environmental, Social, and Governance (ESG) factors in investment strategies has gained importance. Research has shown that investors are increasingly pursuing portfolios which show sustainability targets, and ESG aligned investment has been associated with better performance over the long run [5]. The absence of standardized scoring and data in the ESG is however a challenge to the incorporation of the factors into the automated systems [6].

The Low-code/No-code platform has turned out to be a new wave in software development. According to Gartner, it is estimated that more than 70 percent of new business applications will be built with the use of low-code/no-code technologies by 2025 [7]. They are appealing to organizations with limited technical capacity due to the fact that these platforms allow the quick prototyping and deployment of systems with a limited knowledge of programming. Researchers point out their potential in democratizing access to digital innovation in the financial sector [8] although their use in terms of investor profiling and advising system is not explored sufficiently.

In general, although previous studies show progress in profiling, robo-advisory, and ESG investing have been developed, there is an obvious gap in integrating the discussed aspects

into a low-code/no-code model to develop easily accessible tools to serve international investment advisory.

1.3 Research Gap

Despite the noticeable improvements in the field of investor profiling, robo-advisory, ESG integration, and low-code/no-code technologies, there are still a number of limitations to the implementation of these strategies in international capital markets. The literature has tended to discuss these areas separately and has not suggested a solution that would relate them to each other to make up a viable, accessible system that could be applicable to a small and medium-sized advisory firm.

The following table summarizes the gaps identified in prior work and outlines how this research intends to address them:

Aspect	Existing Literature	Proposed Solution
Investor Profiling	The conventional approaches are founded on fixed questionnaires and theoretical themes like the risk-return framework of Markowitz [1], [2]. Such methods do not reflect the dynamic and individual investor preferences.	Create a profiling module consisting of risk tolerance, ESG preferences, industry focus, and geographic interests interactively via a low-code/no-code interface.
Robo-Advisory Platforms	Research points to their effectiveness and automation [3], [4]. They however tend to be restricted to domestic markets and are heavily coded and infrastructured hence inaccessible to the SMEs.	Create a rule-based recommendation engine on a low-code/no-code framework to decrease the cost of development and open access to smaller companies.
ESG Integration	Studies reveal that ESG has a positive effect on the long-term performance [5], [6]. However, there is poor	Add synthetic ESG scoring to the dataset, so that the ESG is incorporated in customized

	consistency in scoring and lack of data to facilitate integration.	recommendations without losing the transparency.
Technology Adaption	It is reported that they are increasingly used in industries [7], [8]. Its use in financial services is addressed, but its empirical applications to advisory systems are sparse.	Prove that the application of low-code/no-code to the field of financial advisory is real through the creation of the entire system of investor profiling and recommendation.

Table 1: Comparison between systems

1.4 Research Problem

The development of the international capital markets has provided a situation where the investors are faced with thousands of possible instruments whether in equities, commodities, indices and funds. Although this diversity gives the company a chance to diversify its portfolio, it also comes with a lot of complexity in the decision making process. Offering the services of small and medium-sized advisors and individual investors does not always have the skills on how to estimate these instruments, and they may lack the resources to implement sophisticated technological solutions.

Existing investor profiling and recommendation strategies have a number of weaknesses. The conventional profiling instruments are too dependent on fixed questionnaires and the judgment of advisers, which cannot match dynamic investor preferences including changing risk appetite or sustainability factors. This may cause the lack of correspondence between investor expectations and suggested portfolios and the degradation of trust in the advisory process [1], [2].

Robo-advisory solutions seek to automate investment decisions, but on the one hand, they are limited to a particular domestic market, on the other hand, they require considerable technical infrastructure to operate [3], [4]. As a result, these platforms will be locked out to small companies which are not sufficiently large to fund proprietary development or connection to mass infrastructures. According to recent surveys, the use of robo-advisory services has increased significantly in Asia and other emerging markets, but smaller companies have been unable to overcome technological and financial challenges [9].

The growing significance of Environmental, Social, and Governance (ESG) factors has added to the picture further. Although there are existing studies that determine the positive relationship between ESG alignment and long-term performance [5] , [6], no uniform or

standardized system exists on how to integrate ESG into the investor profiling systems. According to reports published in 2022, even though there is an increasing flow of ESG investments, the absence of transparent methodology remains a source of uncertainty to investors and advisors [10].

In the meantime, low-code/no-code technologies have become an option on lowering the obstacles to digital systems development [7], [8]. According to the recent forecasts made by Gartner (2023), more than 80 percent of new business applications are going to be built on low-code/no-code platforms by 2026 [11]. Although this rate of adoption in the industries has been high, their application to the financial advisory systems has been minimal. The evidence of viable applications, which integrate low-code/no-code solutions and investor profiling, ESG integration, and international investment opportunities with each other, into a single framework, is scanty.

That is why, the following research problem can be formulated: *How can a low-code/no-code system be developed and deployed to offer personalized, transparent, and internationally inclusive investment advice by considering ESG factors, and available to small and medium-sized advisory firms?*

2. Objectives

2.1 Main Objective

The overall goal of the research is to design and develop a low-code/no-code Investor Profiling and Recommendation System in International Capital Markets. The system will allow the investors to provide their preferences in terms of risk tolerance, investment timeframe, industry selection, geographic preferences, and ESG priorities, and then create personalized and transparent investment recommendations. The solution is intended to show the relevance of low-code/no-code platform in financial technology, specifically in serving small and medium-sized advisory companies with limited resources to develop advanced systems.

2.2 Specific Objectives

2.2.1 User Requirements

The following are the user requirements of the proposed system:

- Create a user-friendly interface that investors can use to key in profiling data.
- Make it possible to integrate the ESG preferences with the conventional investment criteria.
- Provide categorized recommendations whereby the suggestions are made based on user preferences.
- Provide access to system-generated outputs which are reviewed, interpreted and presented by advisors to clients.
- Basic reporting (exportable summaries) Support: Advisors and support investors with limited reporting capabilities.

2.2.2 Functional Requirements

FR-01

Functional Name	Investor Profiling Module
Input	Investor-specific data such as risk tolerance, investment horizon, industry preference, geographic focus, and ESG priorities.
Process	The system records and validates profiling data through interactive forms, mapping preferences to predefined categories.
Output	A structured investor profile stored in the system for recommendation processing.

FR-02

Functional Name	Investment Database
Input	Curated dataset of approximately 400 instruments including equities, commodities, indices, and funds,

	with attributes such as ticker, industry, region, currency, indicative risk, and ESG score.
Process	The system organizes and maintains data for consistency, enabling efficient querying and filtering by the recommendation engine.
Output	A structured repository of investment options accessible to the recommendation module..

FR-03

Functional Name	Recommendation Engine
Input	Investor profile data and instrument dataset attributes
Process	The system applies a rule-based weighted matching algorithm to align investor preferences with suitable instruments.
Output	A ranked list of recommended investment options with explanatory notes.

FR-04

Functional Name	Dashboard & Reporting Interface
Input	Recommendation outputs generated by the system.
Process	The interface formats and visualizes results for clarity, supporting comparison of alternatives and report generation.
Output	Interactive dashboard view and exportable summary reports for investors and advisors.

2.2.3 Non Functional Requirements

- Usability: The system must possess friendly UI, providing clear workflows both for investors and for advisors.
- Scalability: The system must allow expanding the dataset with additional instruments, without performance degradation.
- Transparency: The system shall prove that it's recommendations are made according to logic understandable to end-users, so that they will be able to trust it.
- Performance: The system must be capable of producing recommendations fast, with minimal time spent processing.
- Accessibility: The system shall be operational through web-based platforms which do not require any high-profile hardware, or installation.
- Maintainability: The system has to be easily modifiable and updatable via the low-code/no-code platform.

3. Methodology

The proposed system will be created based on Agile methodology that is well known to be flexible and iterative in software development. Agile methods allow the process of constantly improving requirements and implementing the feedback given by the users regularly, making sure that the system is changing according to the needs of the investors and the advisory firms. Agile encourages incremental development by the division of the project into sprints, during which planning, development, testing, and review are performed in every cycle. Such a solution reduces risks, enhances flexibility in response to emerging needs, and assists with the provision of system features in due time [12].

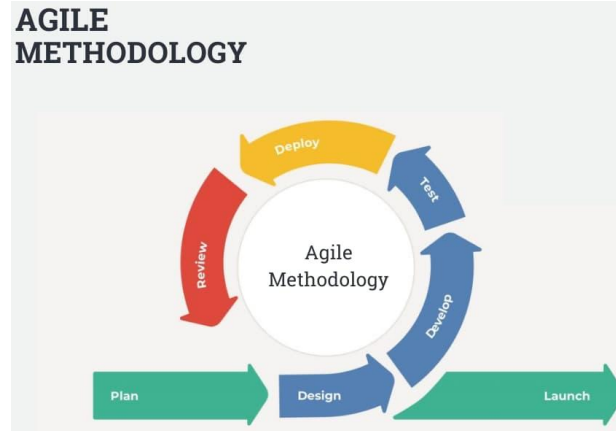


Figure 1: Agile software development life cycle

3.1 Requirement Gathering and Analysis

The identification of requirements will be carried out, considering both the investors and advisory firms because they are the main stakeholders in the system. The key needs that will be identified through literature on investor profiling and robo-advisory systems include risk tolerance measurement, inclusion of ESG preferences, and coverage of multiple assets [3], [5]. As well, advisory use-cases will be examined to make sure that the system meets the reporting and transparency requirements.

The requirements obtained are classified into:

- User Requirements- A straightforward, easy to use profiling system to investors, and clear recommendations.
- Functional Requirement - Profiling modules, dataset management modules, recommendation logic modules, and dashboards.
- Non-Functional Requirements Usability, scalability, maintainability, transparency.

This step wise process allows requirements to be precise, tangible, and executable such that they can be properly designed to allow the system to be designed, and unnecessary ambiguity is avoided in the implementation process [13].

3.2 System Architecture

The system architecture will be in form of facilitating modular integration of important components. It will consist of:

- Investor Profiling Module to capture risk appetite and preferences and ESG considerations.

- Investment Database with close to 400 maintained instruments of stocks, commodities, indices and funds.
- Recommendation Engine is a rule-based weighted algorithm that compares the inputs of investors with those in the database.
- Graphical and Exporter of recommendations Dashboard and Reporting Module.

The architecture is modular in nature whereby individual parts can be extended or substituted without interfering with the entire system. Moreover, scalability and data security will be a design factor so that the system will be able to accommodate larger datasets and safeguard sensitive investor data [14], [15].

3.3 Technical Architectural Design

The technical design will be split into database architecture and web application architecture, which will provide effective management of data and scalability of the system.

- Database Architecture:

The data will be organized in such a way that it makes querying and screening of investment objects effective. Attributes will be stored in tables, which will include ticker, industry, region, currency, risk level and ESG score. Redundancy will be minimized by the use of normalization methods, whereas query performance will be enhanced by indexing. In order to have a robust database, the strategies like replication of databases will be put into consideration to have data available and intact [16].

- Architectural Design: Web Application:

A low-code/no-code tool will be used to implement the system (Bubble.io or Microsoft Power Apps). These systems facilitate modular user interface creation, process automation and database integration. The connectors will be built on the REST and provide external integrations since it might include real-time financial data in the future. The interface will also be usable and responsive in that it will be accessible to various devices [17].

3.4 The Proposed Project's Overall Design

The project will be designed with the user-centered design philosophy, making sure that the profiling, processing, and reporting functions are closely combined. The system will have an easy-to-use interface where investors will feed the inputs in the recommendation engine. The results will be represented in dashboards and exportable to the advisors.

The testing will be continuous and iterative and done at various stages of development to ensure its accuracy, usability, and transparency of the recommendations. Subsequent development cycles will include feedback provided by stakeholders hence enhancing system acceptance and reliability. The design is also such that any future improvement, such as the addition of live financial data, or machine learning algorithms, can be implemented without significant restructuring [18], [19].

4. Technology

The suggested system will be designed based on hybrid technological stack that will guarantee usability, scalability, and adaptability to the requirement of the investors. The choice of the technology is one that helps to strike a balance between the simplicity with which development is achieved and the complexity of financial data that could be processed.

4.1 Front-End

React shall be used to implement the user interface because it is a responsive and component-oriented web application building framework. The option allows it to develop an intuitive profiling form, dashboards, and visualization options that display the recommendations in a convenient way.

4.2 Back-End

Python will be used as the back-end of the system, which will provide flexibility to deal with business logic, data processing, and third-party integration. Python has a specific application in financial algorithm development and can be easily integrated with machine learning libraries to have predictive analytical abilities.

4.3 Database

The data on investor profiles, the curated fund of investment instruments, and system configurations will be stored in a **MySQL** relational database. MySQL is selected due to the strength, scalability, and global financial application. It allows efficient querying of data, which is important in coming up with timely investment recommendations.

4.4 Data Collection

Web scraping will be adopted to fill and refresh the investment data of reliable financial data sources, including Yahoo Finance and Investing.com. This makes sure that there is realistic and current data on the equities, commodities, indices, and funds in the database.

4.5 Recommendation Logic

The recommendation engine will utilize a hybrid model that uses rule based matching and a weighted scoring model. This makes sure that the recommendations are transparent, explainable and those that are suited to the preferences of investors.

Besides scoring using rules, a machine learning prediction core will be considered to make it more personalized. Machine learning models can enhance the precision of the recommendations over time by evaluating previous data and investor trends to facilitate the predictive insights, complementing the deterministic rules.

5. Work Breakdown Chart (WBS)

Work Breakdown Structure illustrates a planning approach where numerous tasks are divided, broken into smaller fragments. In this part, the main responsibilities of the project are allocated in alignment with the various stages of agile methodology. It is significantly simpler to finish asks divided into smaller parts. Additionally, in the image below, the project work breakdown structure is depicted, with each box symbolizing a key project milestone.

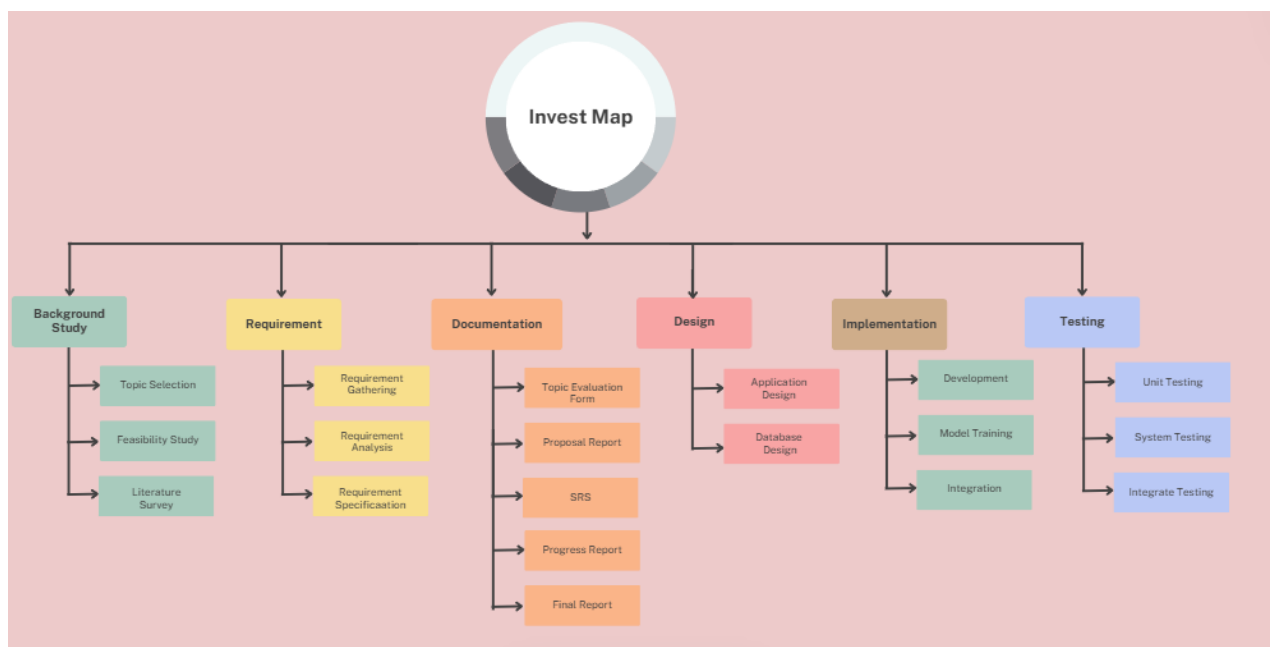


Figure 2: WBS

6. Gantt Chart

The Gantt chart in this section illustrates the project's primary objectives and the timeframe based on agile methodology to fulfil those objectives.

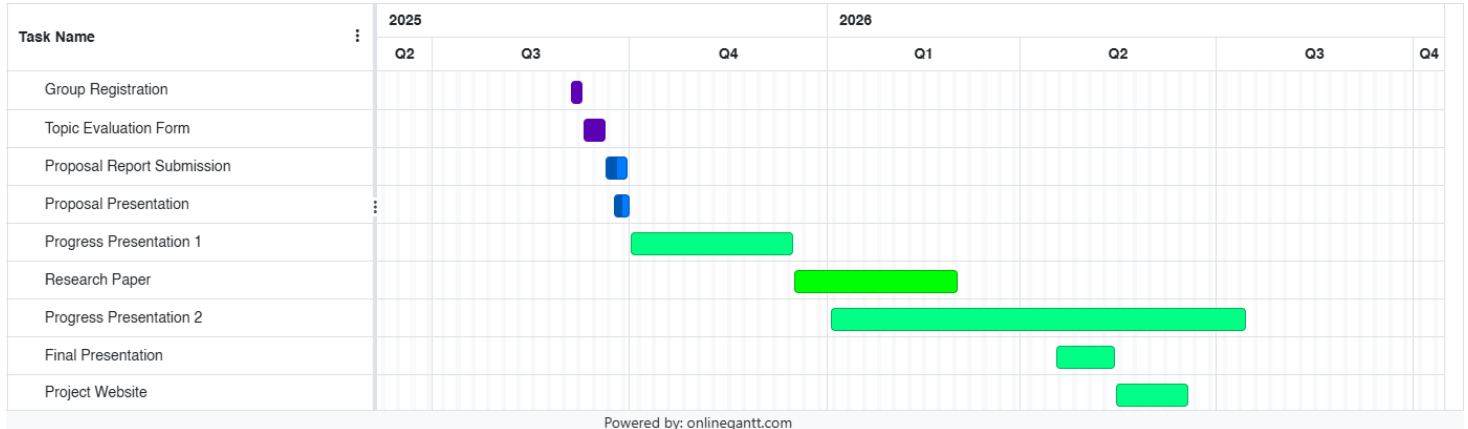


Figure 3: Gantt Chart

7. Commercialization

The suggested system can be highly commercialized in the financial technology (FinTech) sector, specifically among small and medium-sized advisory firms with limited resources to create or acquire costly advisory systems. The system can be used as a more affordable and scalable alternative to regular robo-advisory solutions because of the low-code/no-code technologies.

7.1 Target Market

- **Small and Medium-Sized Advisory Firms:** Firms with small technical skills or that have small budgets which necessitate the use of low cost tools to profile investors and make recommendations.
- **Independent Financial Advisors (IFAs):** Specialists that require digital platforms that are readily accessible to reach out to a wide range of clients.

- Emerging Market Investors: Companies and individuals will be located in developing countries where more developed FinTech solutions are less available.

7.2 Value Proposition

The system will stand out in the market with the following unique selling points:

- Affordability: It is developed using low-code/no-code software, which reduces development and maintenance.
- Accessibility: Web based platform that does not need sophisticated infrastructure.
- Personalization: Progressive profiling with ESG intervention, which is in line with the existing trends of investments in the world.
- Transparency: recommended procedures based on rules and explanatory notes to increase investor confidence.

7.3 Commercial Models

The solution can be commercialized with the help of various business models:

- Software-as-a-Service (SaaS): The licensing of advisory firms on a monthly or annual subscription.
- Freemium Model: Free basic version to individual investors, with extra features (e.g., more advanced reporting, larger datasets, etc.) being sold at a fee.
- White-Label Solutions: Customized system offered to advisory firms under their brand.

7.4 Future Enhancements and Scalability

It can add new datasets, live financial market APIs, and more sophisticated AI recommendations with ease due to the modular system design. In the long run, the platform may be extended to incorporate the aspect of robo-advisory automation, accessibility of the platform via mobile applications, and the ability to connect with portfolio management systems, which will expand its market base.

8. Budget

Item	Estimated Cost (LKR)
Software Tools	100,000
Database and Server Costs	90,000
Machine Learning API Services	88,000
Front-End and Back-End Development	100,000
Testing and Quality Assurance	75,000

Table 2: Budget

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